Hall Ticket Number:

MR-22

Code: B424CM5

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular Examinations, July/August-2024

Object Oriented Programming through Java

(Computer Science and Engineering (AI&ML))

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	List the features of OOPS concepts?	1	1	1	1
1. b)	Define a constructor and list the types of constructors.	1	1	1	1
1. c)	What are the uses of super keyword in Inheritance? Give examples.	1	1	2	1
1. d)	Differentiate between Interface and Abstract class with example?	1	1	2	2
1. e)	What is interthread communication?	1	2	3	2
1. f)	List the different string handling functions?	1	2	3	2
1. g)	List the types of event listeners.	1	1	4	2
1. h)	Discuss the event delegation model used in Java's AWT.	1	2	4	3
1. i)	State the difference between applets and applications?	1	3	5	2
1. j)	Give the subclasses of JButton class.	1	2	5	2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain various types of operators with example.	5	1	1	1
2. b)	With suitable code segments illustrate various uses of 'final' keyword.	5	3	1	2
	OR				
2. c)	What do you understand by type casting & automatic promotion? Explain with suitable examples.	5	2	1	2
2. d)	Demonstrate Nested class concept with an example.	5	3	1	2
3. a)	Give brief description about the various access control mechanisms used in java.	5	3	2	2
3. b)	Illustrate on how packages can be created, defined and accessed.	5	2	2	2
	OR				
3. c)	Develop a Java Program to create an abstract class named Shape that contains two integers and an empty method named printArea(). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method printArea () that prints the area of the given shape.	5	4	2	4
3. d)	What is interface in java? Explain the need of interface mechanism in java?	5	2	2	2
4. a)	With a program illustrate user defined exception handling.	5	2	3	2
4. b)	How to handle multiple catch blocks for a nested try block? Explain with an example.	5	3	3	2
	OR				
4. c)	Demonstrate Auto boxing and Auto unboxing with an example program.	5	3	3	3
4. d)	How to achieve synchronization among threads? Write suitable code.	5	3	3	4

P.T.O.

5. a)	Explain about AWT class Hierarchy?	5	3	4	2
5. b)	Explain the procedure to handle mouse events in Java with an Example program.	5	2	4	2
	OR				
5. c)	Write a java program to design a scientific calculator using AWT.	5	4	4	4
5. d)	What are the various layout managers used in Java?	5	2	4	2
6. a)	What is the role of event listeners in event handling? List the Java event listeners.	5	2	5	2
6. b)	Write an applet to display the mouse cursor position in that applet window.	5	3	5	2
	OR				
6. c)	Explain the life cycle of an applet.	5	2	5	2
6. d)	Why swing components are preferred over AWT components?	5	2	5	3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome